

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 423 — Past Questions & Materials
by Renance

Course: MTH 423
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — MTH423 Numeric Methods 2020 2021 Second Semester

Level 400 • Second Semester • PastQuestion

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UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

. COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

Year '20, SECOND SEMESTER EXAMINATION 2020/2021

Subject: MTH 423 - Numerical Methods

Course Code/Title: MTH 423 - Numerical Methods

4 questions. Time Allowed: 2 Hours. | NOTE: 30 Marks for CoA

Question 1)

a. Let a function $y = 3x - 2$ for $0 \leq x \leq 10$. Construct a forward and a backward difference table (3marks)

b. A function $f(x)$ is given by $f(x) = x^2 + 1$. Find $A^0 f$, (3 marks)

c. Given $y = 1 + xy$; $y(0) = 1$. Compute for $y(0.2)$ correct to 4dps using $h=0.01$. (19.5marks)

Question 2 (6)

a. State 3 types of Linear multi-step methods

b. The concentration of a salt y in a home, made soup

3.5y7. At the initial time $t=0$, the salt concentration in the tank is 5 at $h=1.5$ mm. What is the concentration of the salt at 3 minutes?

Question 3 0

a. Define Interpolation (1mark)

b. Given the table of value as:

is. see \$0 th 3e 4.0 4.5

"a. i... 12.45 15.70 _19.52__ 23.75

formula (8marks)

Find $y(3.34)$ using Newton's Forward difference interpolation

c. Given the table

300. 14.20 (15.65 17.17 18.82 _20.63

| (8.5marks)

Construct a divided difference table

and difference table of the

Taylor's series method at

maker is given as a function of time by $y = 34.5 - 0.004t$. Using Runge-Kutta of order II]

(5)

Question 4 /

- State Modified Euler's iterative formula (2.5 marks)
- Given $dy/dx = xy$ with $y(1) = 5$ with a step size $h = 0.1$ and an interval of $[1, 1.5]$, find the value at $x + h = 0.4$ using Runge-Kutta of order II to a Tol 10^{-6} (15 marks)

Question 5 re)

- If the Chebyshev polynomial of the 1 kind is given by the current iterative formula: $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$, where $T_0(x) = 1$, $T_1(x) = 2x$. Generate the first-four polynomial series using the current iterative relation. (6marks)

2 3 4 a 6

- Given the Maclaurin's series of the form $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$ Use the relation of the 15 kind to economize the error in the series with a Precision Tolerance (Tol 10^{-6}) in a close interval $[0, 1]$ when $x = 1$. (11.5 marks)

Question 6 Wy,

- Use the Modified Euler's iterative formula to solve $y' = 1 - y$; $y(0) = 0$ at $h = 0.1$. (8 marks)
- Apply the Taylor's series method in finding the value of $y(1.1)$ and $y(1.2)$ correct to 3dp given that $y' = 2x - x^2y$; $y(1) = 1$. Take the first three terms of the Taylor's series expansion. = (9.5 marks)